

INTRODUCTION TO PROGRAMMING (CSE141)

Lab Examination – Set C

Max Marks: 40 | Time Allowed: 2 Hours

Submission folder: Name_ERP (e.g., Ali_20261234). Save all four specified .cpp files in this folder.

Exercise 1. Examination Hall Checkerboard 10 marks

An SMCS examination hall has r rows and c columns (each 1 to 10). Its seat map contains 0 for an available seat and 1 for a blocked seat. Print a labelled checkerboard: an available cell has label A if (row + column) is even and B otherwise, where row and column start at 0. Every blocked seat prints # regardless of location.

After the map, print the number of available A seats, the number of available B seats, and the maximum length of a consecutive, horizontally adjacent stretch of AVAILABLE seats in any single row. Blocked seats break a stretch. Use exactly one space between characters in a printed row.

Submission file: hall.cpp

Sample Input — Keyboard input

```
3 5
0 1 0 0 1
0 0 1 0 0
1 0 0 0 0
```

Sample Output — Console output

```
A # A B #
B A # A B
# B A B A
A seats: 6
B seats: 5
Longest open segment: 4
```

Marking rubric

Assessment criterion	Marks
Checkerboard position logic	3
Correct blocking and formatting	2
A/B counting	2
Longest open horizontal segment	3
TOTAL	10

Exercise 2: Run-Length Encoding

[10 marks]

Read a string of letters stored as a C-string. Print its run-length encoding: each run of a repeated character becomes the character followed by its count. Process it **character by character** and print directly.

You are **not allowed** to use any extra array or any built-in string function.

Sample Input

String: aaabbbbcc

Sample Output

a3b4c2

Marking rubric:

Rubric	Points
Reads the string correctly	2
Character-by-character scan, no extra array or built-ins	3
Correct run detection and counting	3
Correct encoded output	2

Submission: rle.cpp

Exercise 3: The Photo Editor (Case Study)

[10 marks]

A photo editor stores an image as a square grid of $n \times n$ brightness values (n up to 8). A user opened an image and clicked one button once; the grid changed from the first picture to the second, shown below. You must read the original image from the user, work out **which single operation the button performed**, and reproduce the edited image.

Study the before and after and decide for yourself whether the button transposed, rotated, or shifted the image, then write the code.

Opened image

```
1 2 3
4 5 6
7 8 9
```

After one click

```
1 4 7
2 5 8
3 6 9
```

Your program reads n and the image, then prints the edited image.

Sample Input

```
3
1 2 3
4 5 6
7 8 9
```

Sample Output

```
1 4 7
2 5 8
3 6 9
```

Marking rubric:

Rubric	Points
Reads n and the image into a 2D static array correctly	2
Identifies the correct operation (transpose) from the example	3
Correct nested-loop implementation	4
Correct image output	1

Submission: photo.cpp

Exercise 4: Sales File Study

[10 marks]

A file `sales.txt` holds an unknown number of integer daily sales figures, one or more per line (at most 100).

- Read all figures into a static array.
- Using the original order, find the longest run of consecutive days where sales increased over the previous day, and display its length.
- Then arrange all figures from lowest to highest using **insertion sort** (one step of the task) and write the ordered list plus the total to `salesreport.txt`.

If the file cannot be opened, print an error message and stop.

Sample Input

```
sales.txt:
10 20 15
18 25 30
```

Sample Output

```
Longest increasing run: 3
Total: 118
(salesreport.txt written)
```

Marking rubric:

Rubric	Points
Opens file with error check; reads unknown count into array	2
Correct longest-increasing-run logic on original order	3
Correct insertion sort (ascending) as a sub-step	3
Writes ordered list and total to output file	2

Submission: salesstudy.cpp